

JUN WEI WANG

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I am a Software Engineering student in the Integrated Circuits track at Fudan University, with an interdisciplinary focus that connects computer science, hardware systems, and applied ML. I like to learn by building: from low-level systems and hardware-adjacent tools to high-performance software and physics-based simulations, and I am especially interested in how software, chips, and ML models come together to solve real-world problems efficiently. Recently, I have been involved in undergraduate research, competition teams, and industry-oriented projects around computer systems, hardware acceleration, and ML-driven tools, as well as mentoring peers and helping organize technical activities.

Before coming to Fudan, I received an offer to study in the Mathematical and Physical Sciences program in the Faculty of Arts & Science at the University of Toronto (St. George campus), which affirmed my academic preparation in mathematics, physics, and computing. I decided to study in China—and specifically at Fudan—because I want to reconnect with the language and culture and work in a tech ecosystem that is fast-moving and hands-on. My goal is to grow into an engineer who can contribute to projects in China while staying connected with the global tech community.

Outside of academics, I enjoy violin, photography, and hardware/IoT tinkering, as well as building side projects in full-stack development and exploring new tools and programming languages. These hobbies keep me curious, creative, and comfortable working across both the abstract and the practical sides of engineering.

Education

Fudan University, Shanghai, China 2022 - present

- Software Engineering
- Interdisciplinary in Integrated Circuits (集成电路)
- Chinese Government Scholarship, CSC Scholarship Category A

Tongji University, Shanghai, China 2021 - 2022

- Pre-university Chinese (预科)
- CSC Scholarship Category A
- Top scoring CSC pre-university graduate

Elgin Park Secondary, Surrey, British Columbia, Canada 2018 - 2021
Grades 10 - 12

- Tiger Sun Memorial Computer Science Scholarship
 - Exceptional computer science student & entering university to study computer-related courses
- 2020/21 BC Achievement Scholarship
 - One of 8,000 graduates with top academic achievement in British Columbia

Nisga' a Elementary Secondary School, New Aiyansh 2017 - 2018
Canada
Grade 9

Technical Skills

- **Programming:** Python; C/C++ (Qt; Basic OpenGL; drivers/system-level; embedded on Arduino/ESP32; CUDA Platform); JavaScript (TypeScript); Java; Rust
- **Full-stack developer:** End-to-end web apps; integration across frontend, backend, APIs, and data stores; familiar with customizing JS compilers (Webpack, SWC, Rollup, etc.); SPA/SSR/ISR solutions;
 - **Frontend:** React/Next.js; TypeScript; TailwindCSS; common UI libraries
 - **Backend:** Node.js with Express; REST/GraphQL APIs; PostgreSQL/MySQL; Redis; common backend frameworks
- **Cloud & DevOps:** AWS (EC2/S3; static/backend hosting); Aliyun (ECS/OSS; hosting); GitHub CI/CD basics

Project Experience

Compass HLS DSE Application 2025 - present
PI — FDUROP (Fudan Undergraduate Research Opportunities Program)

- GUI for Compass HGB0-DSE (Hierarchical GNN + Bayesian Opt) to auto-generate HLS pragmas
- Led VS Code plugin + Svelte front-end; integrated AST-based pragma synthesis and Vitis HLS backend
- Custom server backend
- Custom Python inference backend
- Future work includes integration of a newer model and optimizations to PPA prediction depending on FPGA architecture

Sichain Semiconductor 2025 - present
MOSFET manufacturing — testing & QC pipeline

- ML-aided AOI (ResNet-18/50) for wafer/assembly defect detection
- Rust + React (Tauri) tooling (with ThreeJS for rendering) to ingest and streamline QC stages across fabs/assembly (including CP, WLBI, AOI, etc.)

Jarvis (E-bike CMS) 2025 - present
ESP32S3 BLE/Wi-Fi/4G central management/monitor system

- Master/multi-slave BLE, SoftAP + web UI, LTE/IoT telemetry, GPS tracking, anti-theft/unlock; integrates Far-driver ESC + AntBMS; ESP-IDF + Svelte framework embedded front-end

Loong Squad 2024 - 2025
Superscalar OoO (out-of-order) LoongArch32 CPU (Chisel)

- LoongCore: superscalar OoO CPU (LoongArch32) with branch prediction, caches, pipeline flush handling; Scala/Chisel + SBT
- Team prize: 2nd Place, Loongson Cup group contest (Loong Squad@FDUPPI);
WeChat post: WeChat
- Results: prelim 1st place, finals 6th overall; ~5.3x speedup on provided tests at ~100 MHz

- Passed FuncTest/PerfTest/SysTest and boots Linux on LoongSoC

DNN Acceleration on FPGA 2024

Hand-written number classification (MNIST) CNN on FPGA (PL) with Linux PS + Ethernet I/O

- Implemented a CNN in PL logic; PS side bridged to PL and exposed Ethernet for data in/out between FPGA and external devices
- Built a custom Python client over Ethernet; delivered live webcam handwritten-digit recognition pipeline;

Full-stack Web Platforms 2024 - present

Next.js + DO + MySQL + Cloudflare + S3

- Built and hosted multiple in-progress apps with custom domains: glorialan.com (Next.js), backend, and jimmys-cms
- DigitalOcean hosting with managed MySQL; domains via GoDaddy; Cloudflare DNS/CDN; static assets on Amazon S3

TeaPl Compiler 2024

Course project: Custom compiler (built with C++)

- Built front to back: lexer/parser, AST, IR, optimizations, ARM assembly codegen, verification for teaching language **TeaPl**
- TeaPL language spec: teapl.md

DSP BiTCN Post-EQ 2024

ML post-equalization for 64QAM-CAP

- BiTCN-based post-equalizer for 64QAM-CAP signals; references bidirectional TCN-based nonlinear compensation in underwater VLC; midterm DSP project

LoongArch32 Verilator Debugger 2023

GDB-like Verilog CPU debugger (C++/Verilator)

- Built a gdb-style CLI debugger (fdb) for LoongArch32 labs using Verilator & C++; supports step/run/break/print on bin images
- Make/IMG-driven flow for swapping binaries

FDE-CLI 2025

Rust CLI tool for FDP3P7 (SMIMS) board (Fudan's in-house FPGA board for FPGA Principles & Applications course)

- Drives SMIMS engine: program bitstreams, R/W registers & FIFOs; multithreaded CLI with debugging helpers; coursework artifacts in fde-project

Single-Cycle MIPS CPU 2023

Course project: Educational RISC pipeline

- Implemented single-cycle MIPS CPU for coursework/labs
- Supported instructions: add, sub, and, or, slt, addi, andi, ori, slti, sw, lw, j, nop

Work Experience

Fysics / 飞捷科思 (fysics.net) 2024-09 - present
Intern — SDF algorithms, CUDA acceleration, & software development

- In-house physics simulation engine with an OpenGL + ImGui frontend
- SDF algorithms and related pipeline improvements
- GPU acceleration with CUDA
- Software development across data, algorithms, and app integration

English Tutor

2023-09 - present

Private instructor

- 1:1 tutoring focused on English reading, writing, and conversational fluency
- Designed individualized study plans and practice materials

Freelance (Fiverr)

2021 - 2022

Frontend & CS projects; tutoring

- Built React/TS frontends and small apps; custom dashboards and UI refactors
- Tutored CS topics (data structures/algorithms) and tech interview prep; project support

Achievements

- 2025 National College Student Computer Systems Ability Contest — CPU Design (Loongson Cup)
 - Team name: Loong Squad@FDUPPI
 - Second Place
- Mathematical Content in Modeling (MCM) - 2025
 - MCM Problem C (data insights)
 - Honorable Mention
- 2024 National College Student Computer Systems Ability Contest — CPU Design (Loongson Cup)
 - Individual competition
- Fudan University 4th Classic English Speech & Reading Contest — Second Place — 2024.05.18
- Advanced Placements (AP)
 - Calculus BC (5/5)
 - AP Computer Science A (5/5)
- Canadian Computing Competition | University of Waterloo
 - 2020 Senior
 - 2019 Junior

Software Copyrights

- AI-assisted High-Level Synthesis Optimization GUI Tool (Compass) V1.0;
Reg. No. 2025SR2155133; Certificate No. 软著登字第 16811331 号;
Rightsholder: Fudan University; Granted 2025-11-05

Activity Experience

HGBO-DSE demo @ FPGA Education Workshop

2025-07

Co-presenter with advisor

- Presented “HGBO-DSE: Hierarchical GNN and Bayesian Optimization based HLS Design Space Exploration” alongside faculty at the 大模型赋能 FPGA 教育研讨会
- Demoed the HGBO-DSE algorithm and the Compass GUI (HLS DSE)

Programming Club (founder) & 3D Printing Club (co-founder)

- Launched the school's first programming club (Gr. 11-12); taught C++/Java; prepared peers for Waterloo CCC Junior; weekly labs filled the CS room
- Led 3D printing club: modeled parts, ran on-site printing service, trained students on 3D workflow, printers, and troubleshooting

Hobbies and Interests

- **Violin:** ~9 years
 - The Royal Conservatory of Music (RCM): Level 4, level 8
 - Vancouver Academy of Music (VAM)
 - VAMIS (Intermediate Symphony; first violin)
 - VAMSO (Symphony Orchestra; performing major symphonic works at a pre-professional level; second violin)
 - Surrey Youth Orchestra (second violin)
- **Photography:** Street/travel and landscape; exhibitions
 - Tongji University 2022 International Student Photography Exhibition (作品展)
- **Hardware & IoT tinkering:** simple circuit design & soldering; Arduino projects; Raspberry Pi projects; ESP32 wireless smart home modules; Home Assistant; custom 433 MHz Arduino smart home nodes (MySensors); home server/networking; etc.
- **Full-stack web dev:** multiple web apps; React/Vue/Next; Firebase/MongoDB/DigitalOcean/AWS; self-hosted servers and own my own domain name
- **Programming, reverse engineering, hacking:** love to learn about low-level tinkering, tooling, and security/vulnerability related topics

UofT Admission Letter

Congratulations

WE ARE DELIGHTED TO OFFER YOU A PLACE AT
THE UNIVERSITY OF TORONTO.

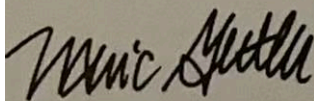
JUN WEI WANG

STUDIES IN MATHEMATICAL AND PHYSICAL SCIENCES
FACULTY OF ARTS AND SCIENCE
ST. GEORGE CAMPUS
UNIVERSITY COLLEGE

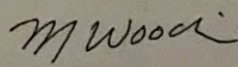
Your record of achievement has earned you a place
in our exceptional community of scholars.

By accepting your offer of admission, you join some
of the brightest minds from around the globe who share a passion
for learning, a deep curiosity about the world, and
a commitment to making a difference.

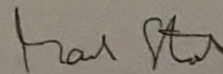
Once again, congratulations. We hope to see you
at the University of Toronto.



MERIC S. GERTLER
President



MELANIE WOODIN
Dean



MARKUS STOCK
Principal

